Lyft vs Uber:
Demand Patterns, Revenue Management, and Competitive Pricing Strategy

OM386 Demand and Pricing Analytics - Project

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Roadmap

Introduction

Market context and the pricing dynamics

Section 1: Demand Elasticity

Establishing baseline ride volume variance across temporal, geographical, and environmental factors

Section 2: Revenue Maximization

Comparing surge frequency and missed opportunities

Section 3: Competitive Landscape

Assessing independent pricing strategies between Uber and Lyft

Conclusion

Key findings and strategic implications for ride-sharing platforms

The Two-Sided Marketplace

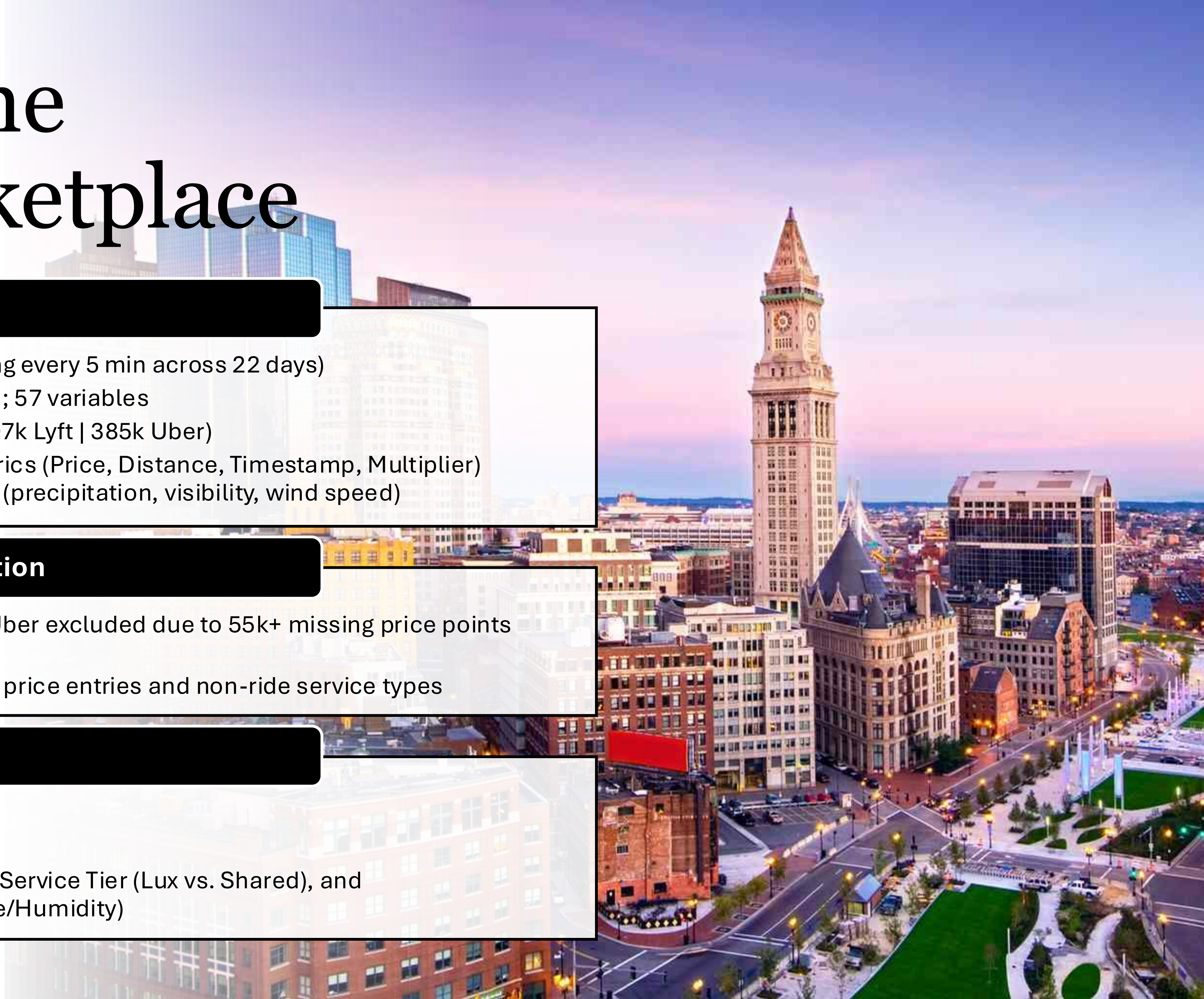
Platforms continuously balance driver supply and rider demand across micro-markets.

The Yield Paradox: High prices suppress demand; low prices fail to attract supply.

Market Efficiency: Surge pricing prevents "wild goose chase" equilibrium, where throughput collapses because drivers spend excess time searching.

Uber/Lyft commissions (20-25%) make revenue a direct function of both volume and fare level.

Dataset: The Boston Marketplace



Data Scope & Composition

- **Timeframe:** Nov-Dec 2018 (sampling every 5 min across 22 days)
- **Raw Volume:** 693,071 observations; 57 variables
- **Analysis Volume:** 637,976 rides (307k Lyft | 385k Uber)
- **Attributes:** Individual trip-level metrics (Price, Distance, Timestamp, Multiplier) merged with real-time weather data (precipitation, visibility, wind speed)

Strategic Filtering & Segmentation

- **Platform Isolation:** Lyft data only. Uber excluded due to 55k+ missing price points and stagnant surge multipliers
- **Quality Control:** Removed all "null" price entries and non-ride service types

Key Analytical Drivers

- **Target:** Price (Total Fare)
- **Primary Lever:** Surge Multiplier
- **Controls:** Normalized for Distance, Service Tier (Lux vs. Shared), and Environmental Factors (Temperature/Humidity)

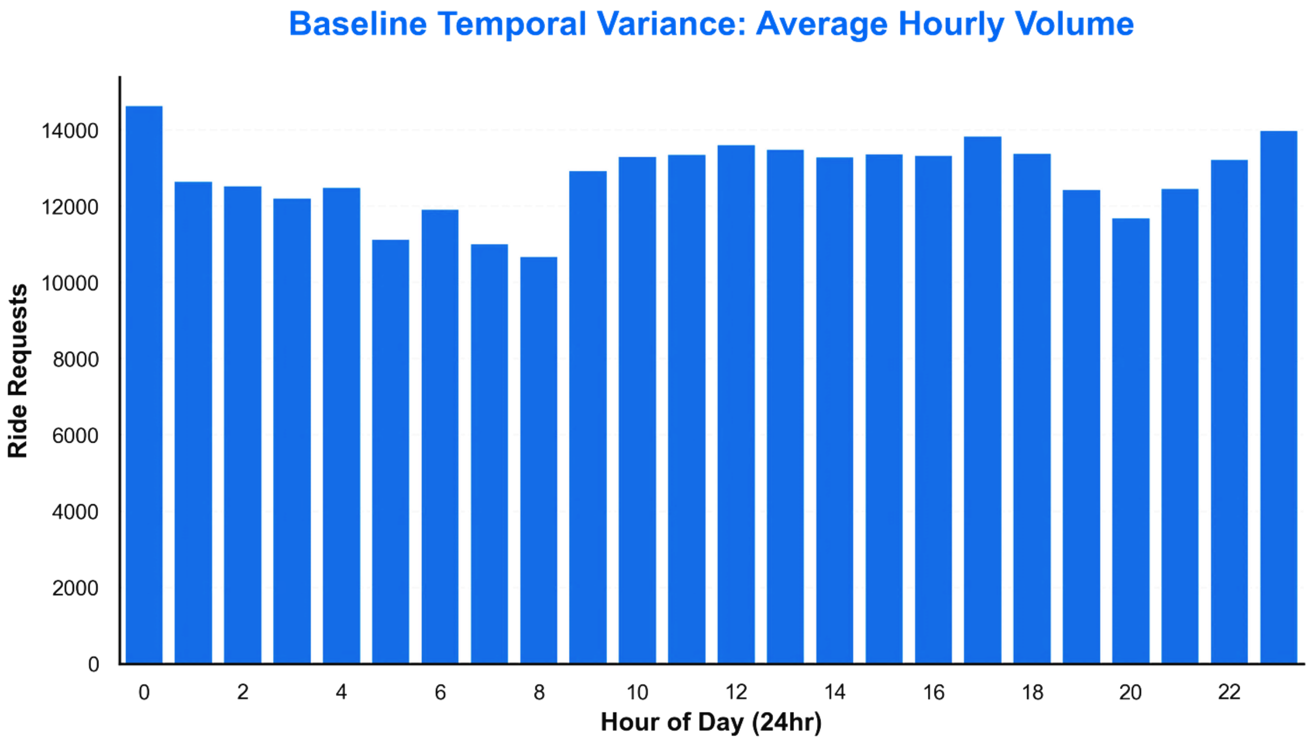
Section 1: Demand Elasticity

How elastic is the demand of Lyft ridership?

$$\ln(Volume) = \beta_0 + \beta_1(price) + \ldots + \beta_n(x_n) + \epsilon$$

Employ Log-Log regression to directly measure demand elasticity.

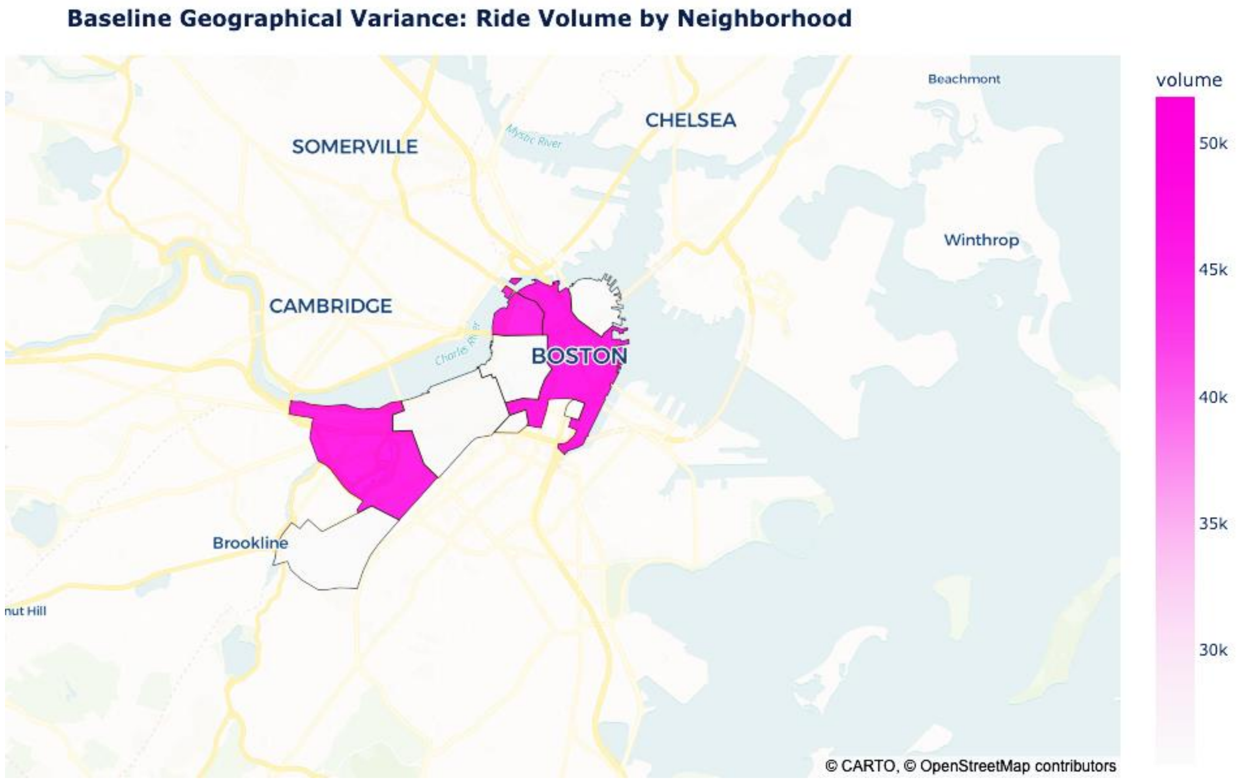
Temporal Variation



Ridership exhibits cyclical peaks during working hours (8am-5pm) and late night social hours (10pm-1am). Though we do note some inconsistencies in the data.

Demonstrates predictable seasonality of demand.

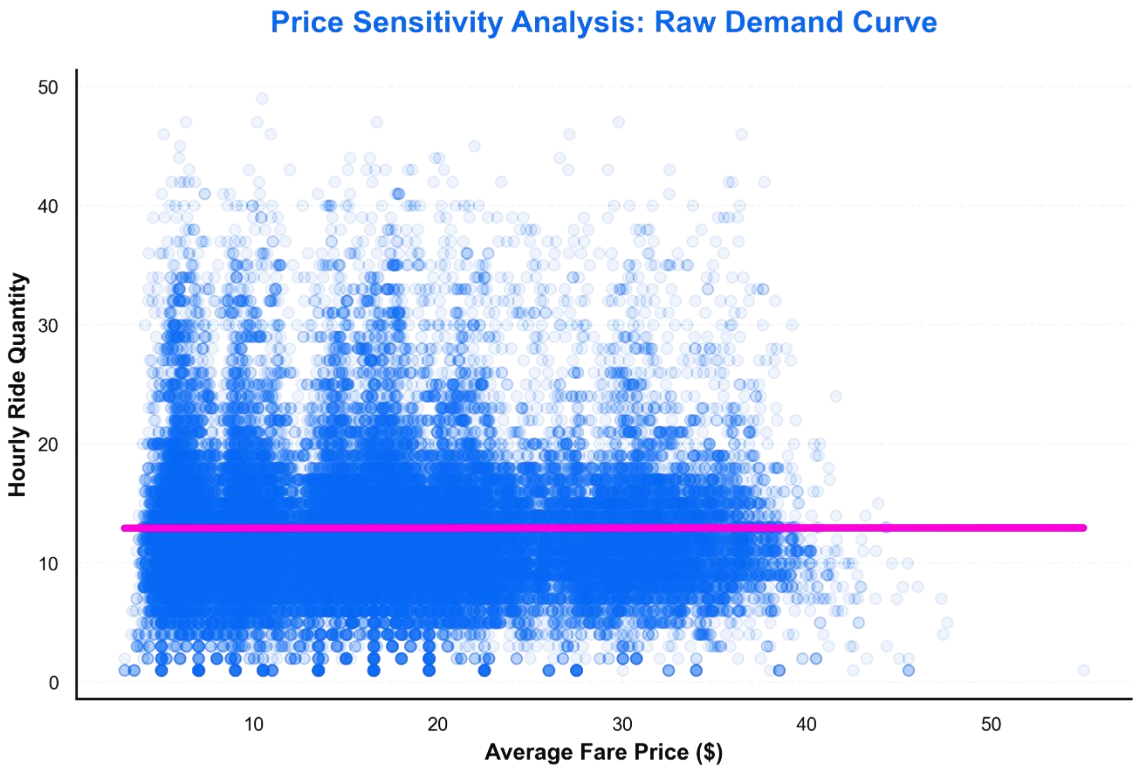
Geographical Variation



Downtown, West End, and Fenway neighborhoods carry a significant portion of the Boston area Lyft demand.

Identifies particularly insensitive regions that consistently exhibit high demand (academic, corporate, & entertainment hubs).

Visualizing the Demand Curve



The price v. demand plot exhibiting relative invariance over increased prices suggests that Lyft riders may be indifferent to price (at least within about the $5-$35 range)

Price and quantity move in lockstep, flattening the demand curve and demonstrating inelasticity.

Measuring Elasticity

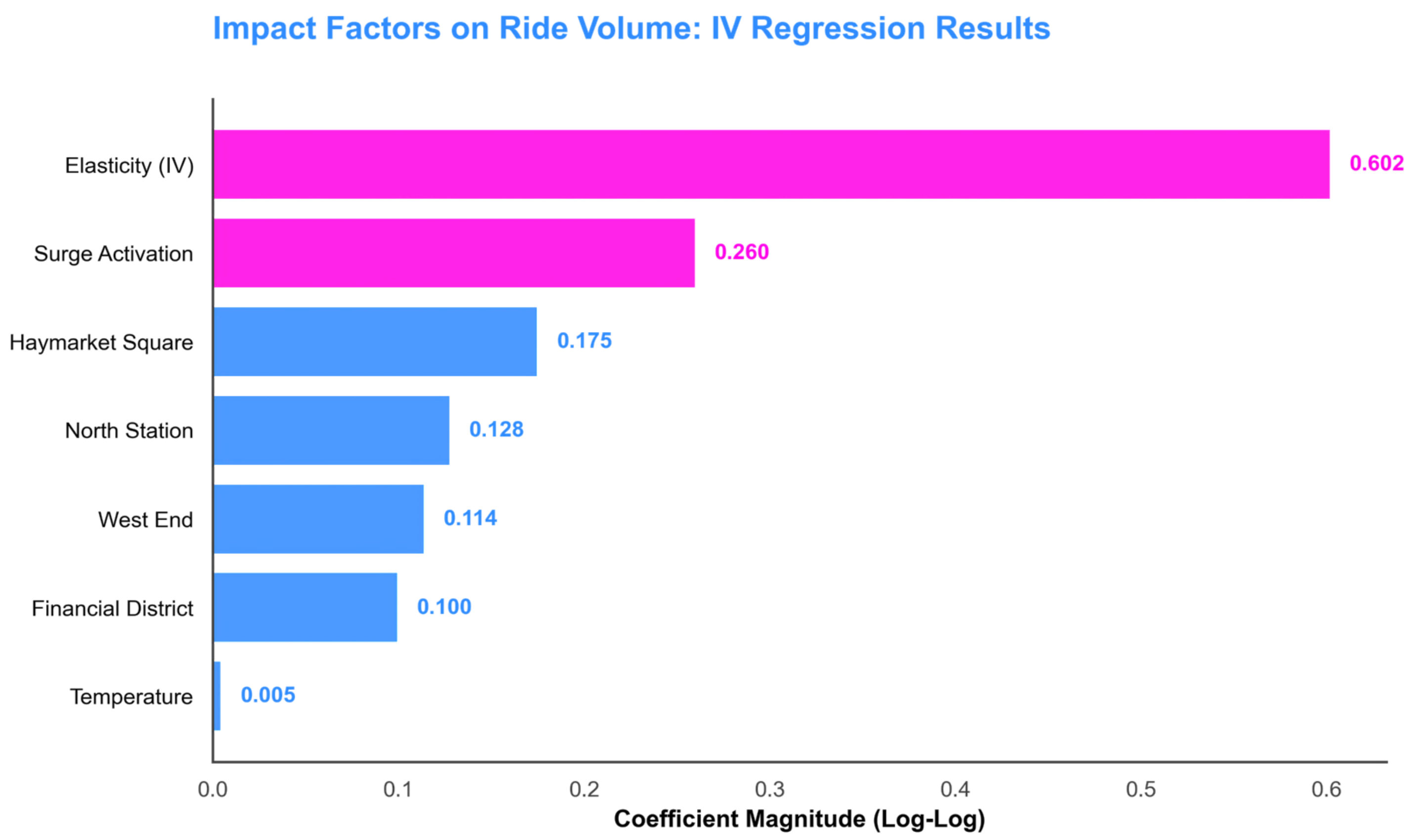
Challenge of simultaneity bias: High Demand $\Rightarrow$ Higher Prices

Using Log-Log form: Allows for direct interpretation of elasticity

Instrumenting lagged price: Cleaner signal of price sensitivity

Controlling surge activation: Measures sensitivity under surge

Log-Log Regression: *Surge-Controlled*



What Matters:

1. Surge controlled IV model yields elasticity of 0.6, demonstrating volume continues to trend with price increases
2. Surge weight of 0.26 means the demand intercepts jumps 26% when surge is activated.

Lyft ridership demonstrates clear demand inelasticity and even jumps as prices are surged. High-volume regions are extreme examples of this.

Section 2: Revenue Maximization

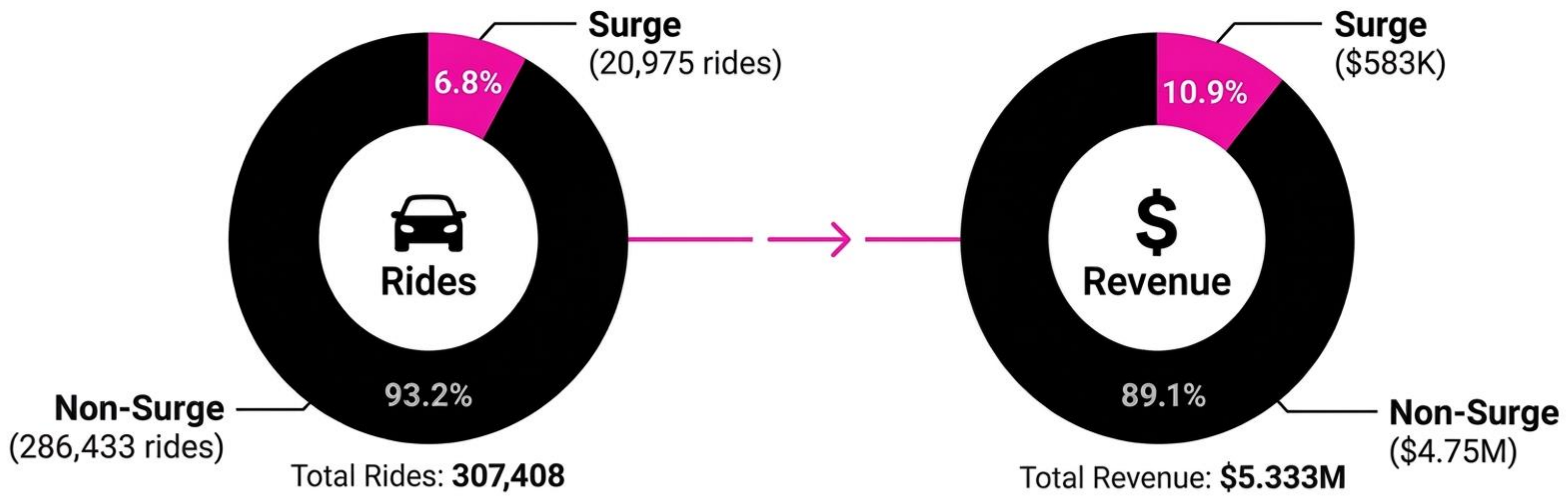
Is Lyft's surge pricing algorithm maximizing revenue, or is the platform leaving money on the table?

$$\text{Max} \sum_{i=1}^{n} P_i \times D(P_i)$$

where P is price and D(P) is the demand function

The surge multiplier is the primary dynamic lever to balance supply and demand.

Surge is a tiny fraction of rides, but a *massive* driver of revenue



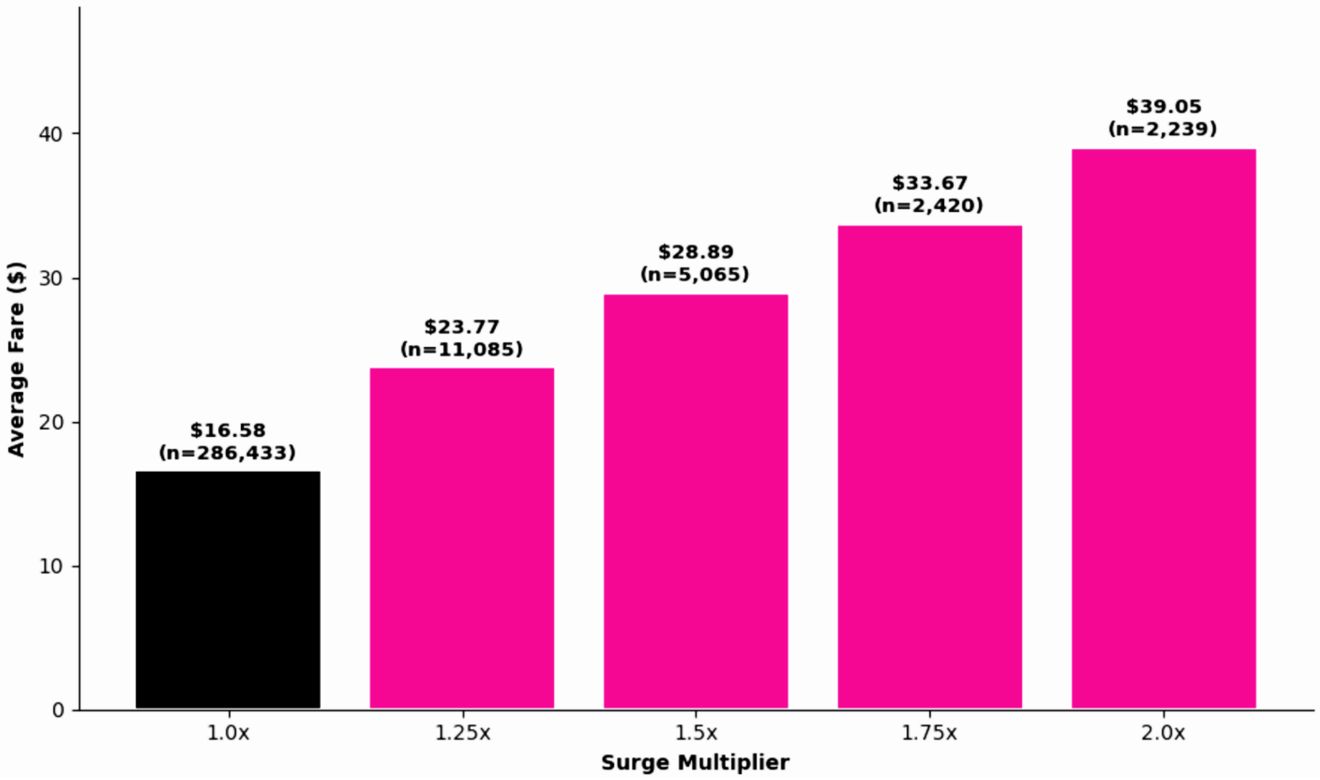
$16.58
Base Fare

$27.84
Surge Fare

+67.9%
Price premium

Surge rides command a 67.9% price premium. However, the 6.8% activation frequency suggests massive untapped potential.

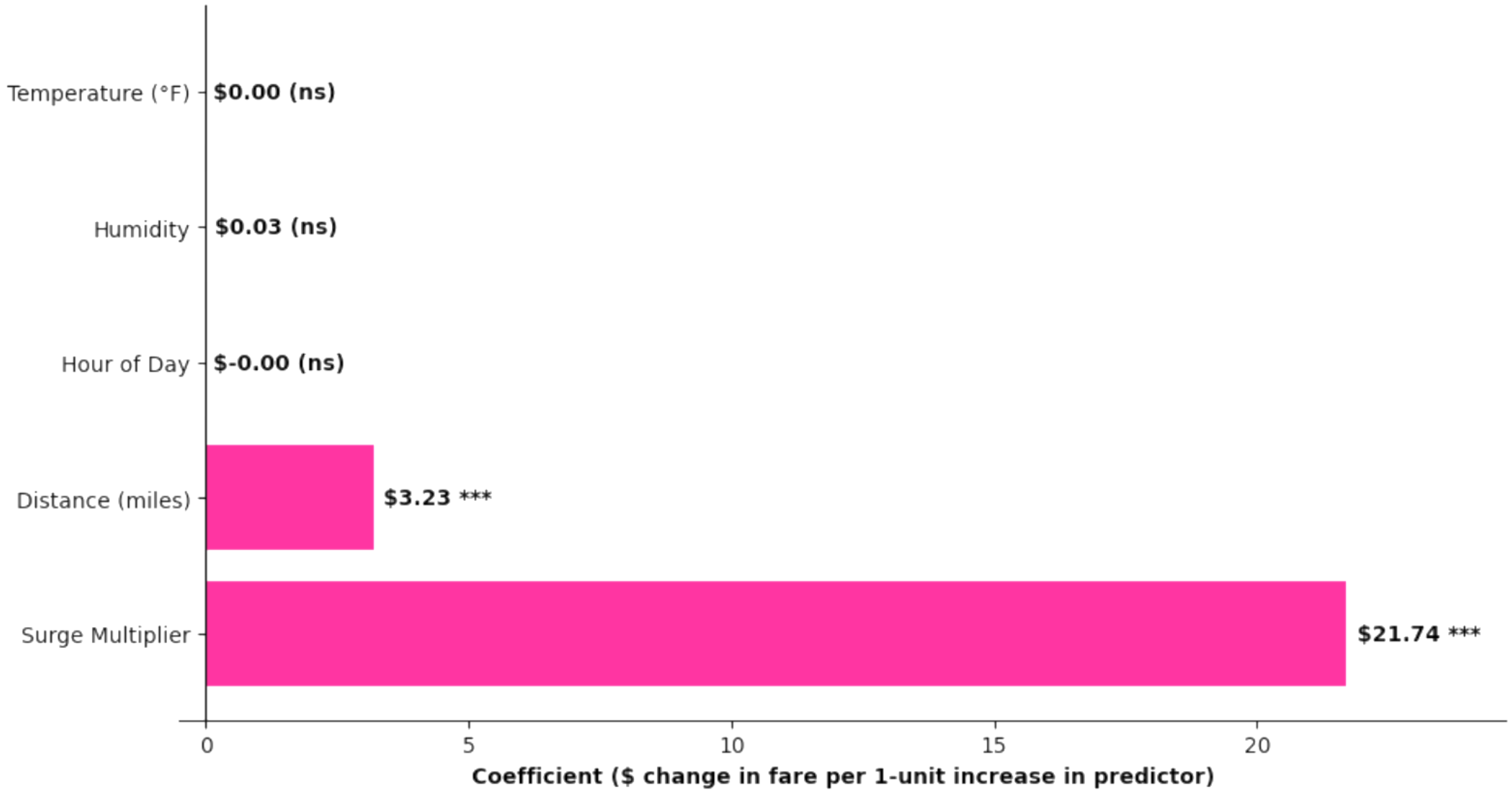
Riders aren't flinching: Revenue grows *linearly* with multipliers



Every 0.25-unit step adds $5.50-$6.00 to average fare. The linear growth suggests demand is not yet exhausted at current multiplier levels.

We are underestimating the consumer's "willingness-to-pay" in urgent scenarios.

OLS Regression: *Fare Drivers*

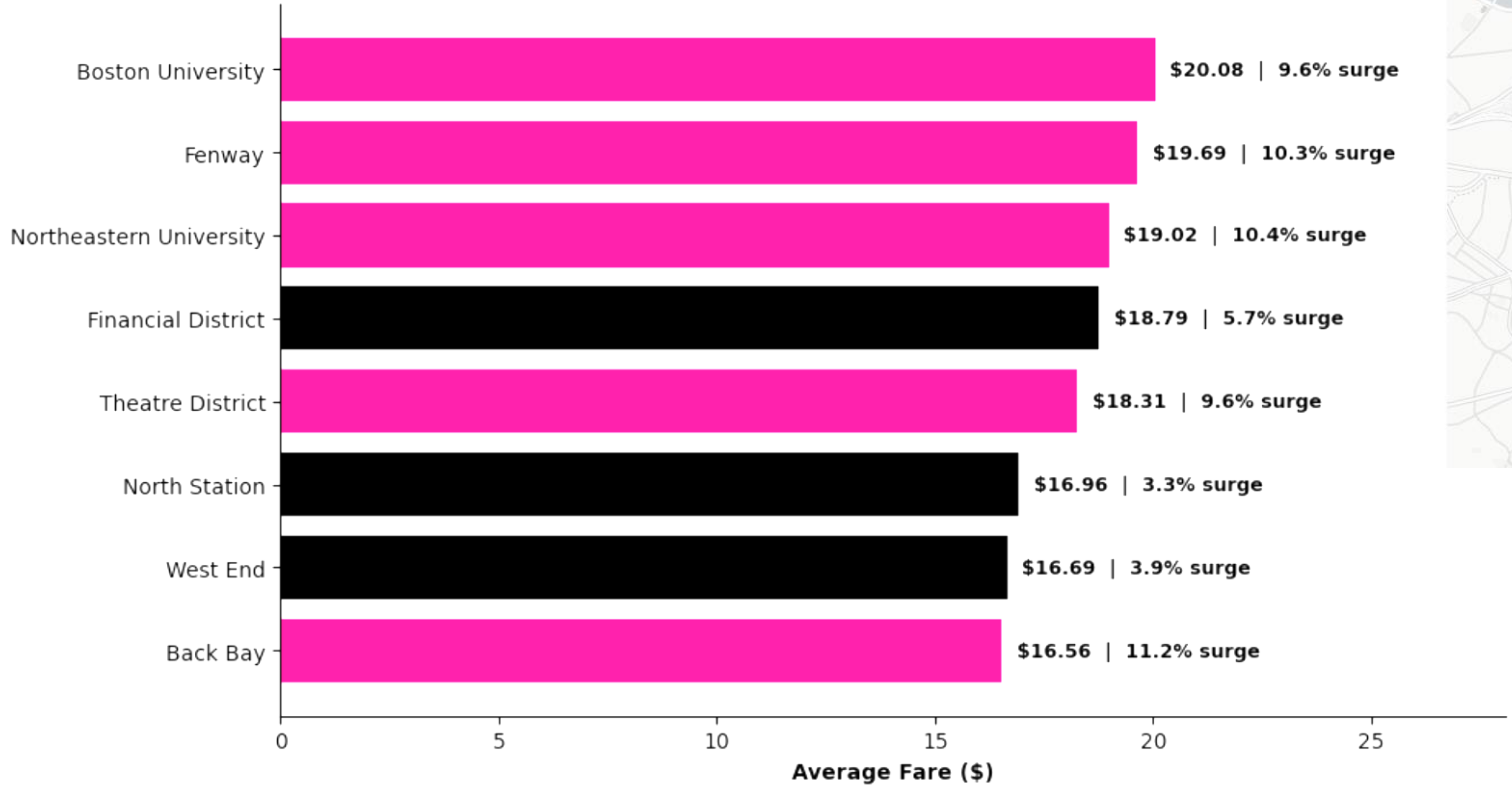


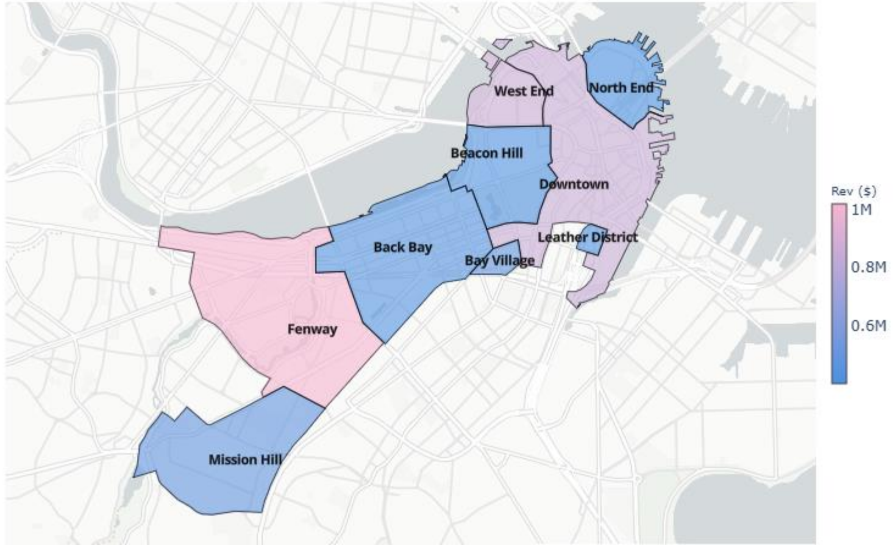
What Matters: Surge multiplier (+$21.74 per unit) and Distance (+$3.23 per mile) are the dominant drivers.

The Blind Spot: Highly predictable external demand signals, specifically Time of Day and Weather, proved statistically insignificant.

Lyft's algorithm is purely reactive. It waits for queues to deplete rather than pricing ahead of known demand spikes.

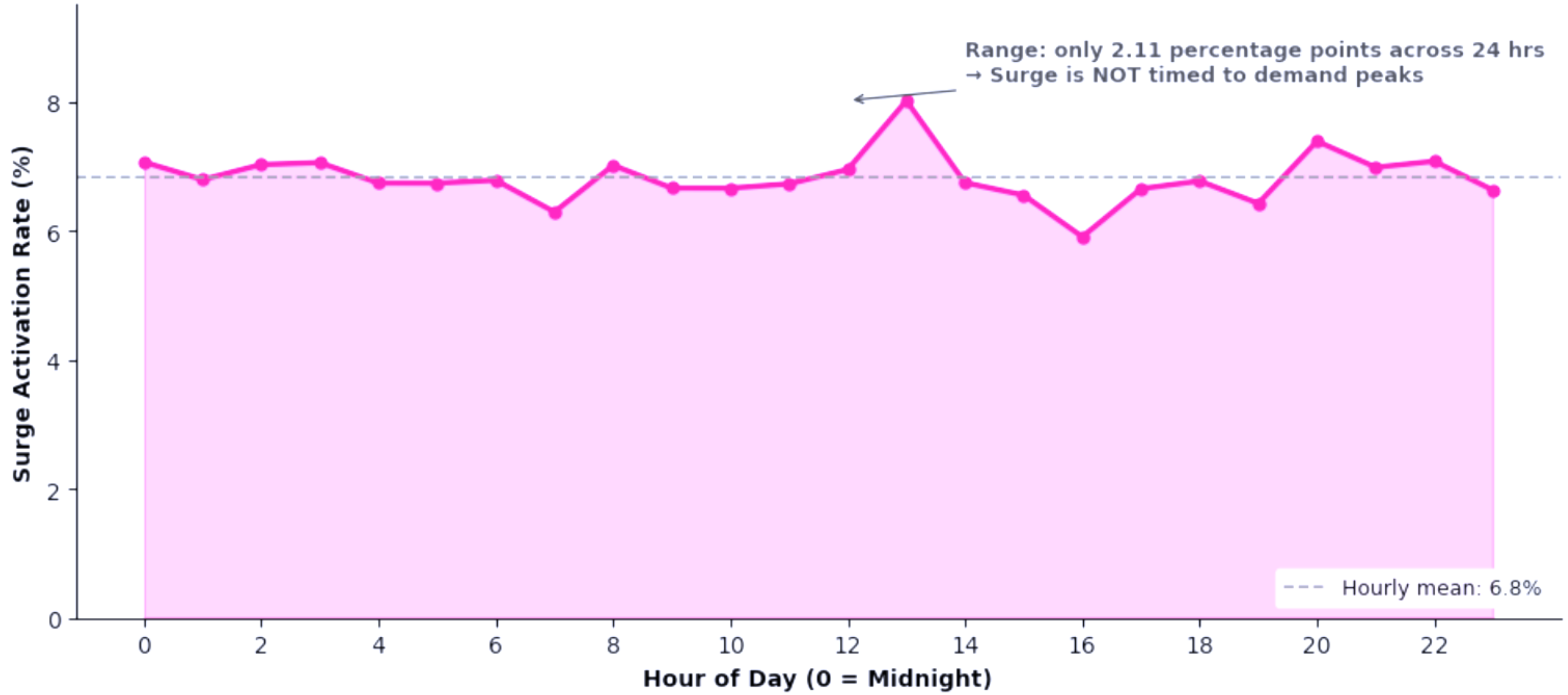
Geographic *Hotspots*: Economic Epicenters vs. Revenue Deserts





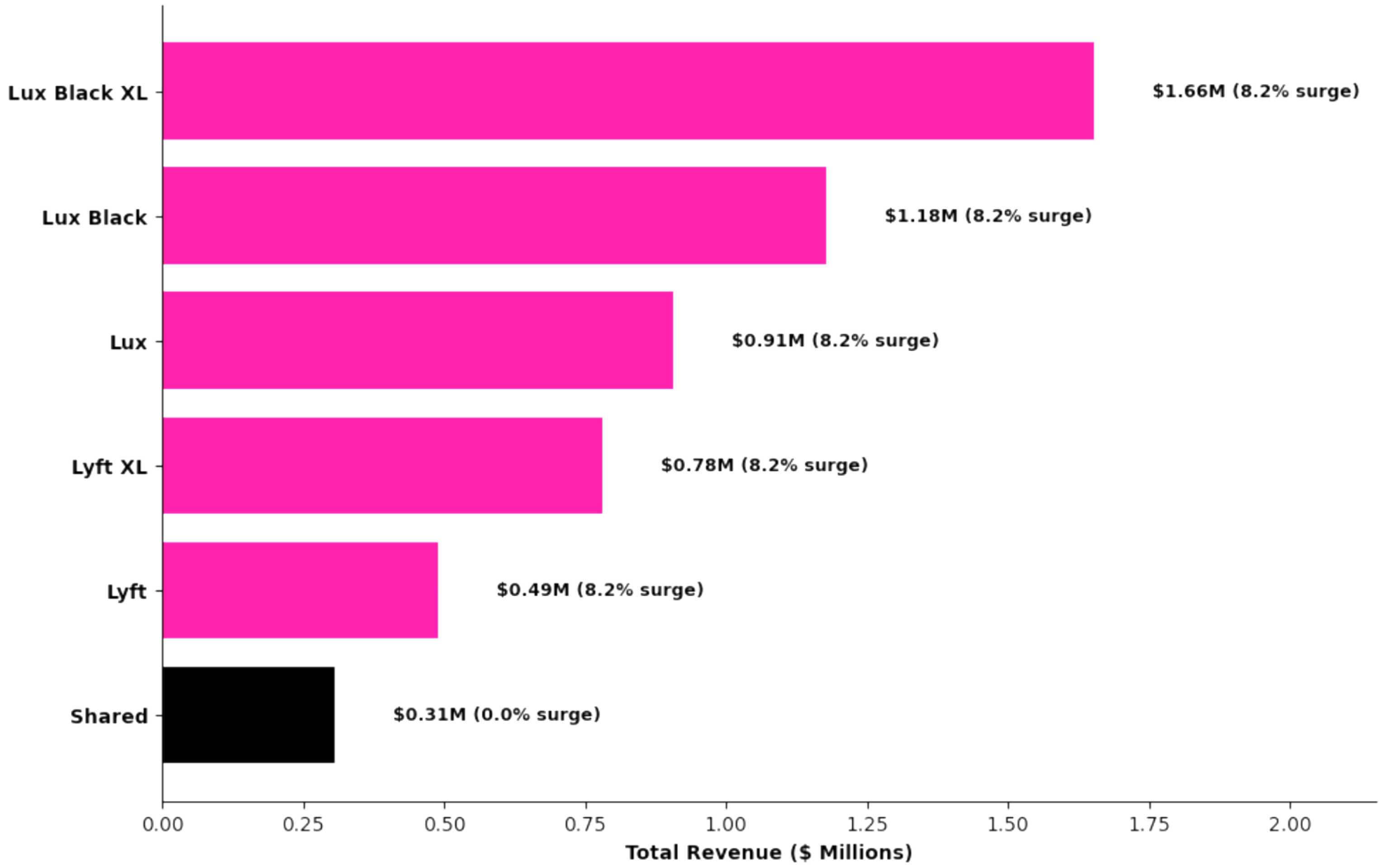
The algorithm successfully identifies spatial hotspots. Surge is applied heavily around major venues and campuses.

Our pricing is blind to *predictable* demand spikes



*The surge rate stays **flat** (approx 7%) all day. It doesn't care if it's rush hour or 3 AM.*

We treat luxury riders like budget riders, *leaving surplus* on the table



8.2%

Identical surge frequency across all premium tiers

Premium riders have lower price elasticity, yet Lyft fails to capture the consumer surplus available in these luxury segments.

Section 3: Competitive Landscape

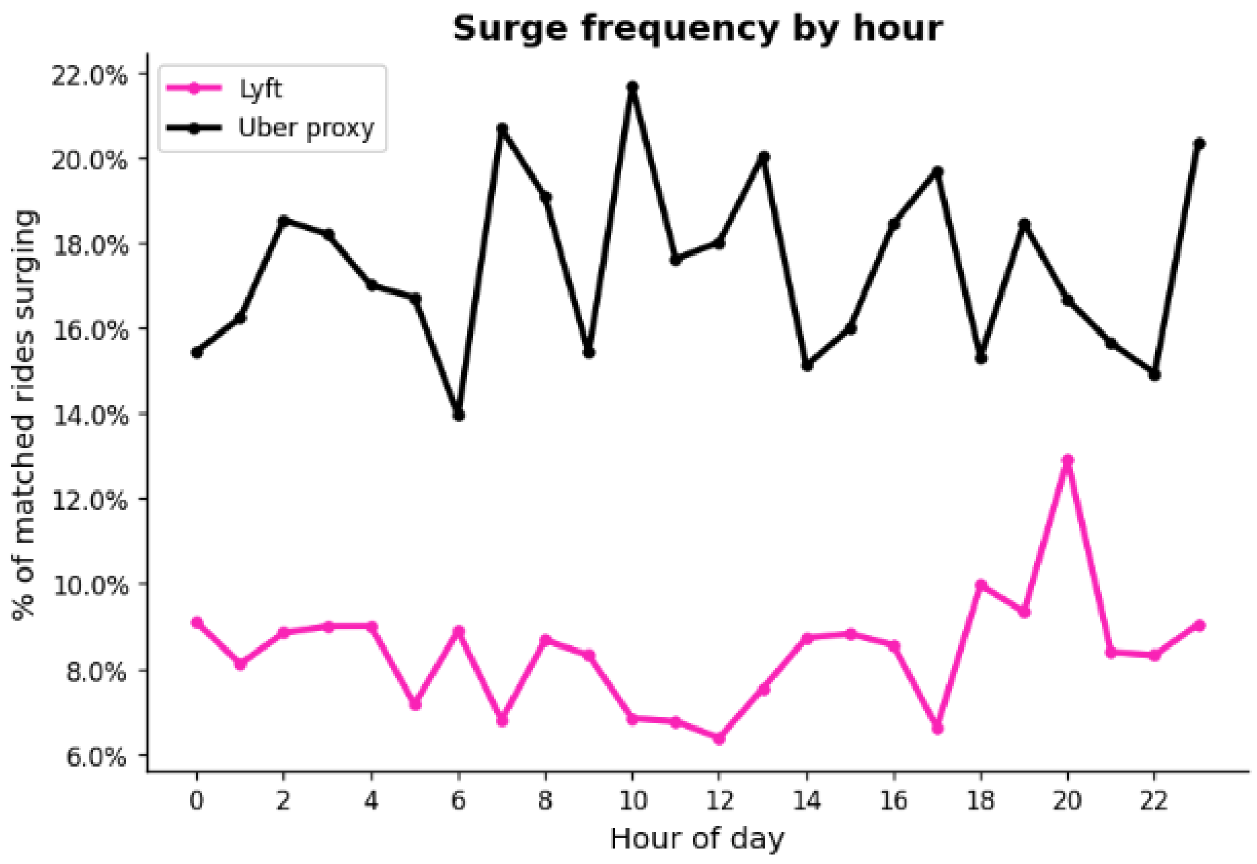
When Uber confirms demand is elevated, does Lyft follow?

Uber surges twice as often, on the *exact same routes*

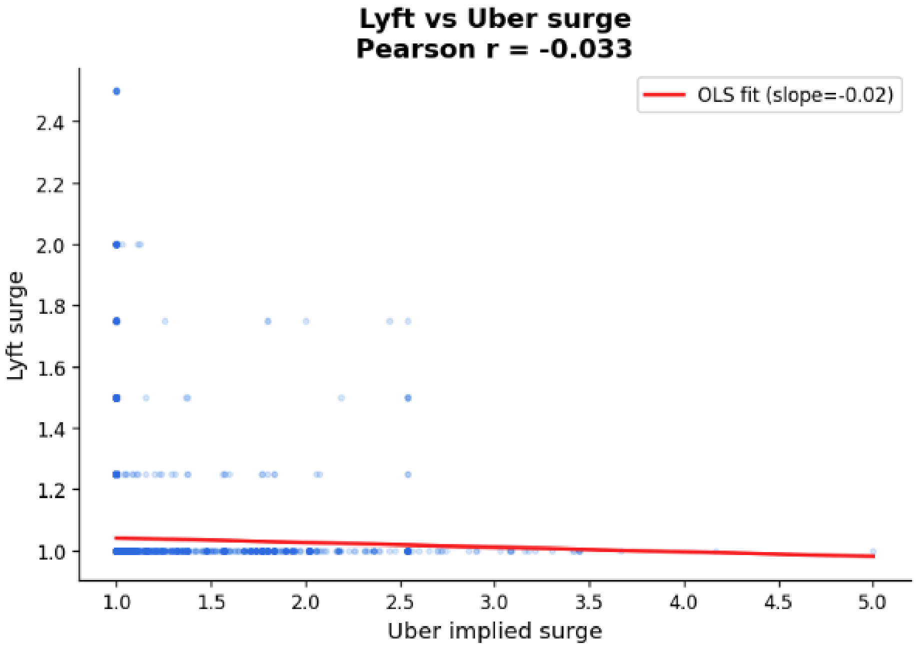
17.5%
Uber surge frequency

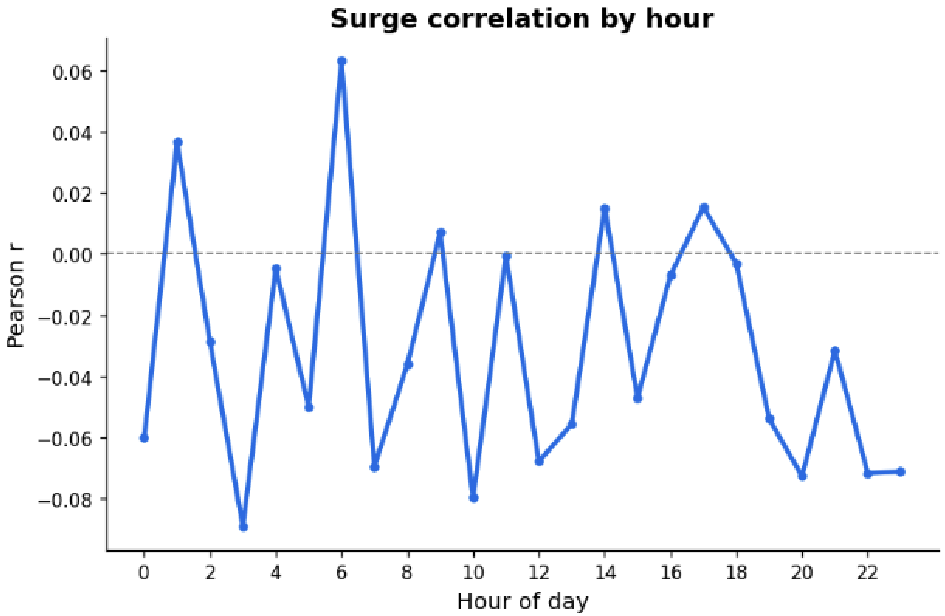
8.4%
Lyft surge frequency

11,834
matched route-time pairs



Lyft does not follow Uber’s surge signals





$r = -0.033$

surge correlation

$\Delta R^2 = 0$

Uber surge adds to OLS model

$p = 0.36$

Uber surge coefficient (OLS)

No lead-lag in either direction

Granger causality test

Does knowing Uber’s past surge help predict Lyft’s next move?

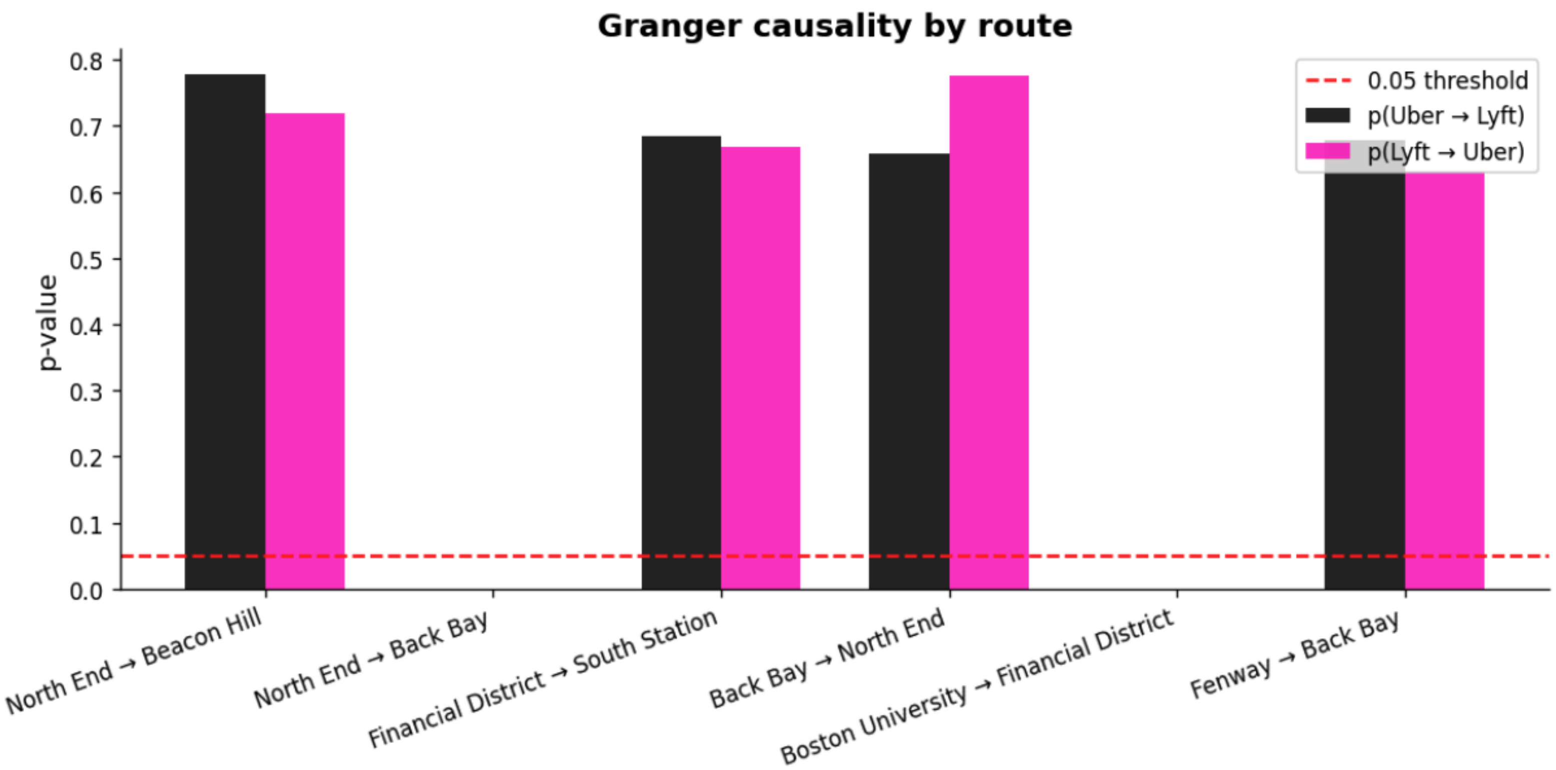
6 / 6 routes

classified Independent in both direction

p > 0.60

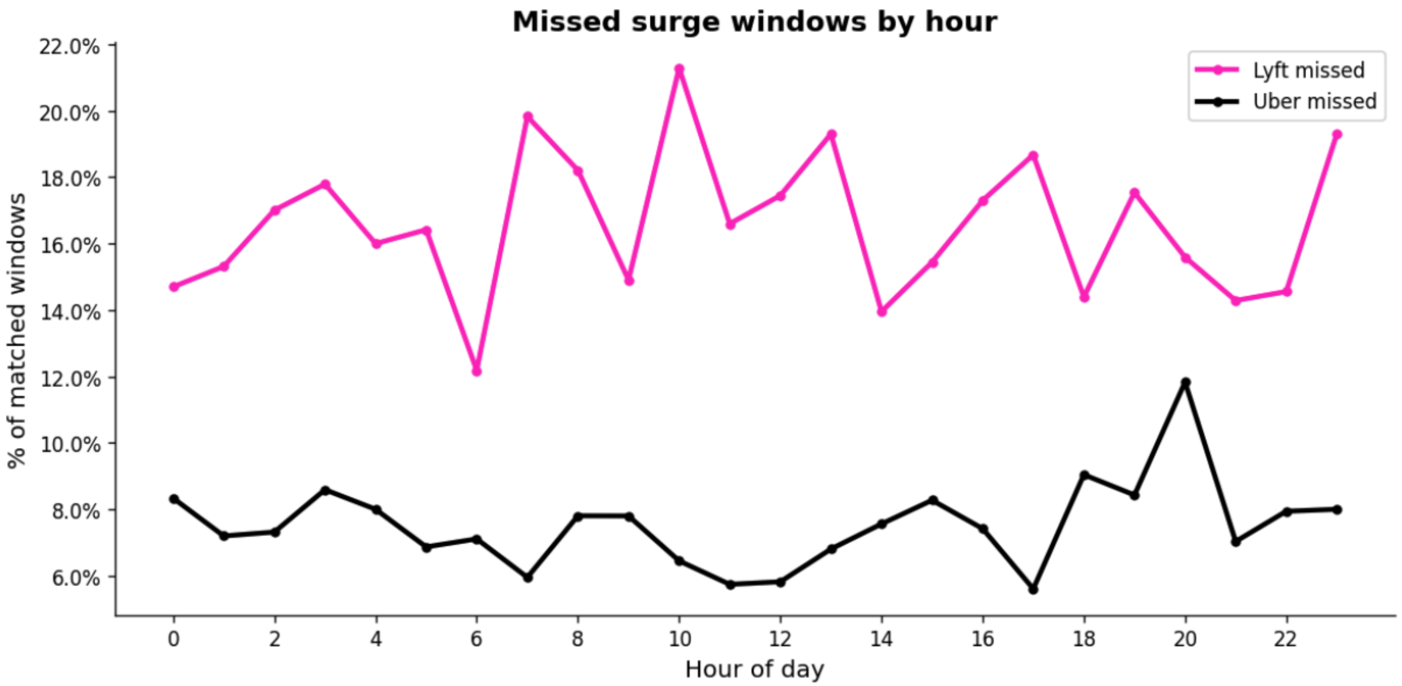
on every route, in both directions

Uber doesn’t lead Lyft. Lyft doesn’t lead Uber. They price in silos.



All bars far above the red 0.05 threshold. Uber’s past surge has zero predictive power for Lyft’s next price move.

Uber's surge is a *free demand signal*. Lyft isn't reading it.



16.6%
windows Lyft misses

1.6x
avg Uber surge in those windows

$1.10
avg fare gap per missed ride

7.5%
windows Uber misses (2x less)

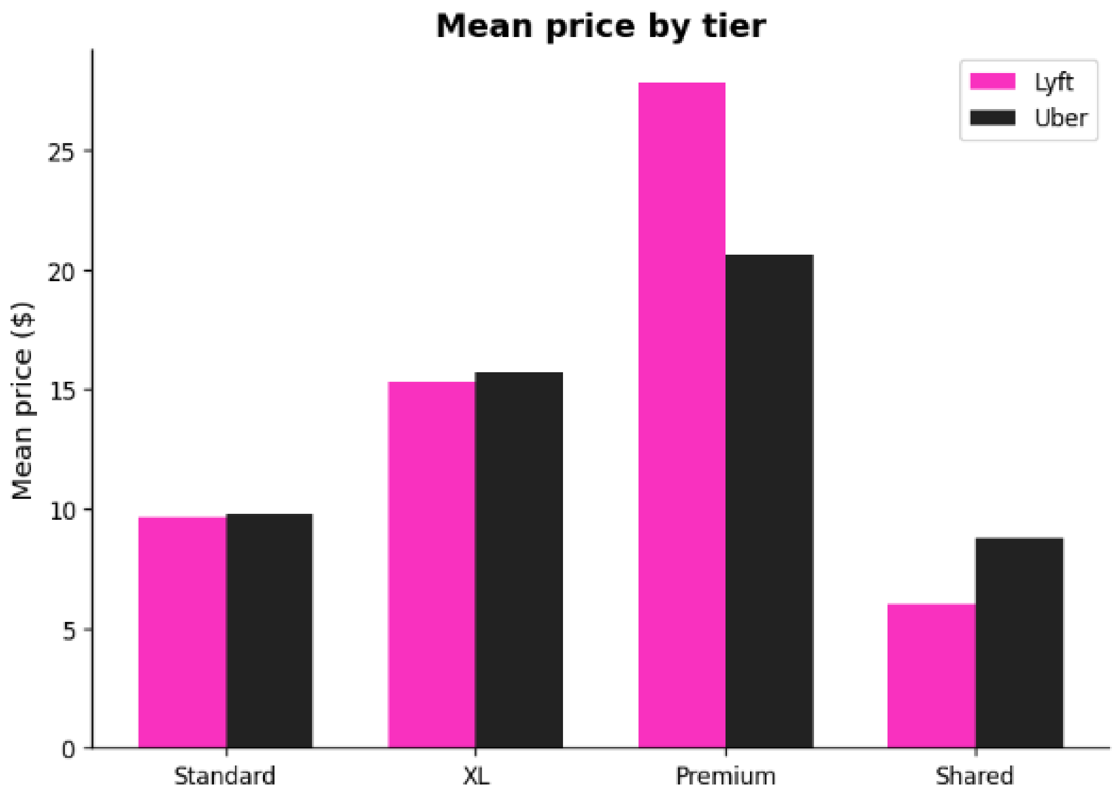
Premium is the *biggest missed opportunity*

Premium tier gap

Uber surge: 1.675x

Lyft surge: 1.038x

On avg $27 rides. Lowest elasticity segment.



Conclusion

Demand is inelastic, revenue grows linearly with surge, and competitors confirm demand, yet Lyft surges half as often as it should.

01

Predictive Triggers

Incorporate historical time/weather data to surge at the onset of rush hour.

02

Tier Differentiation

Apply higher surge multipliers to Lux/Premium tiers where sensitivity is lower.

03

Market Expansion

The 2.0x ceiling is arbitrary. Test higher caps in low-elasticity neighborhoods like Fenway.

04

Competitive Yield Cue

Use Uber's surge as secondary input; in 16.6% of windows Uber is already confirming demand that lyft ignores

Thank you!
Q&A